volcano, near Gargantua's horizon, the gravitational energy has grown hugely negative and overwhelms the centrifugal energy, so the surface plunges downward (Figure 27.5). The critical orbit is on the volcano's rim.

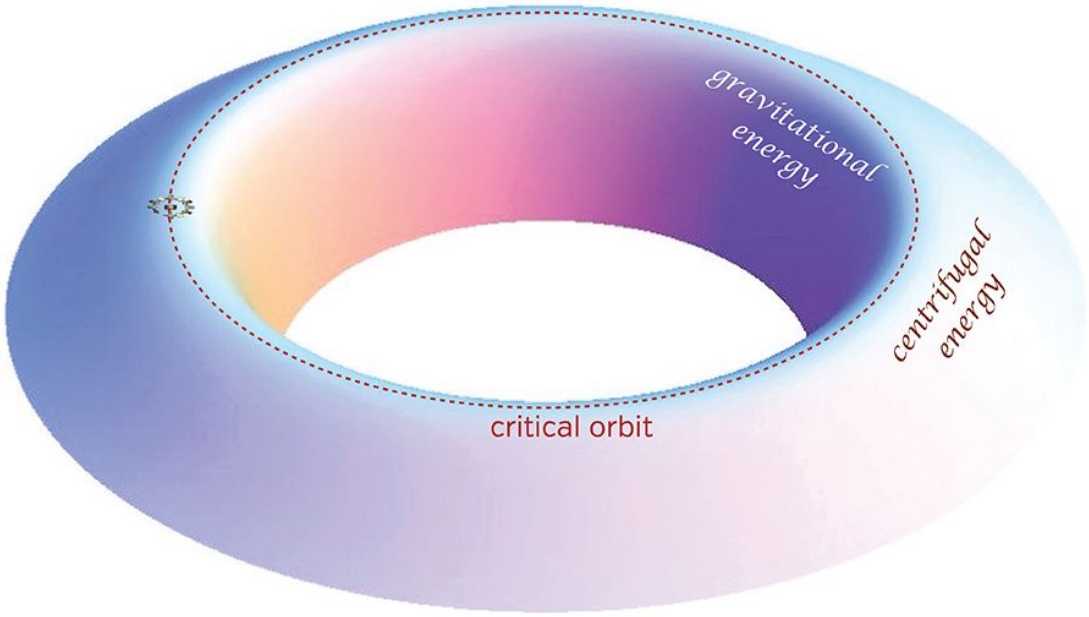


Fig. 27.5. The *Endurance*'s critical orbit on the rim of the volcano, with centrifugal energy and force dominating outside the rim and gravitational energy and force dominating inside. *[Image of the* Endurance *is from* Interstellar.*]*

The Critical Orbit: Balance of Centrifugal and Gravitational Forces

Upon reaching the volcano's rim, the *Endurance*, ideally, would travel around and around it, at constant speed. Because it moves neither inward nor outward, the inward pull of gravity on the rim must precisely be counterbalanced by the outward centrifugal force that arises from the ship's fast circumferential motion.

This indeed is the case, as shown in Figure 27.6—an analog of the force balance plot for Miller's planet (Figure 17.2). At the *Endurance*'s